

Data Dictionary – ICU Mortality Prediction (First 72 Hours)

Identifiers (Relational Keys – Not Modeling Features)

Variable	Description	Source Table
subject_id	Unique patient identifier	patients
hadm_id	Unique hospital admission identifier	admissions
stay_id	Unique ICU stay identifier (unit of analysis)	icustays
itemid	Measurement identifier code	chartevents / labevents / outputevents
charttime	Timestamp of measurement	chartevents / labevents

Outcome Variable

Variable	Description	Source Table
hospital_expire_flag	In-hospital mortality (0=survived, 1=died)	admissions

Demographic Variables

Variable	Description	Unit	Source
age	Age at ICU admission	Years	patients
gender	Biological sex	Categorical	patients
race	Self-reported race	Categorical	admissions
admission_location	Location prior to hospital admission	Categorical	admissions
first_careunit	First ICU unit admitted to	Categorical	icustays
los	ICU length of stay	Days	icustays

Vital Signs (First 72 Hours – All Variants)

Variable	Description	Unit	Aggregation
min_hr	Minimum Heart Rate	bpm	Minimum
mean_hr	Mean Heart Rate	bpm	Average
max_hr	Maximum Heart Rate	bpm	Maximum
min_map	Minimum Mean Arterial Pressure	mmHg	Minimum
mean_map	Mean Mean Arterial Pressure	mmHg	Average
max_map	Maximum Mean Arterial Pressure	mmHg	Maximum
min_sbp	Minimum Systolic Blood Pressure	mmHg	Minimum
mean_sbp	Mean Systolic Blood Pressure	mmHg	Average
max_sbp	Maximum Systolic Blood Pressure	mmHg	Maximum
min_dbp	Minimum Diastolic Blood Pressure	mmHg	Minimum
mean_dbp	Mean Diastolic Blood Pressure	mmHg	Average
max_dbp	Maximum Diastolic Blood Pressure	mmHg	Maximum
min_rr	Minimum Respiratory Rate	breaths/min	Minimum
mean_rr	Mean Respiratory Rate	breaths/min	Average
max_rr	Maximum Respiratory Rate	breaths/min	Maximum
min_spo2	Minimum Oxygen Saturation	%	Minimum
mean_spo2	Mean Oxygen Saturation	%	Average
max_spo2	Maximum Oxygen Saturation	%	Maximum
mean_temp_c	Mean Body Temperature (converted to °C)	°C	Average
max_fio2	Maximum Fraction of Inspired Oxygen	Fraction (0–1)	Maximum
max_pao2	Maximum Arterial Oxygen Pressure	mmHg	Maximum

Laboratory Variables (First 72 Hours – All Variants)

Variable	Description	Unit	Aggregation
min_creatinine	Minimum Serum Creatinine	mg/dL	Minimum
mean_creatinine	Mean Serum Creatinine	mg/dL	Average
max_creatinine	Maximum Serum Creatinine	mg/dL	Maximum
min_lactate	Minimum Serum Lactate	mmol/L	Minimum
mean_lactate	Mean Serum Lactate	mmol/L	Average
max_lactate	Maximum Serum Lactate	mmol/L	Maximum
min_bilirubin	Minimum Total Bilirubin	mg/dL	Minimum
mean_bilirubin	Mean Total Bilirubin	mg/dL	Average
max_bilirubin	Maximum Total Bilirubin	mg/dL	Maximum
min_wbc	Minimum White Blood Cell Count	K/uL	Minimum
mean_wbc	Mean White Blood Cell Count	K/uL	Average
max_wbc	Maximum White Blood Cell Count	K/uL	Maximum
min_hemog	Minimum Hemoglobin	g/dL	Minimum
mean_hemog	Mean Hemoglobin	g/dL	Average
max_hemog	Maximum Hemoglobin	g/dL	Maximum
min_sodium	Minimum Sodium	mmol/L	Minimum
mean_sodium	Mean Sodium	mmol/L	Average
max_sodium	Maximum Sodium	mmol/L	Maximum
min_potassium	Minimum Potassium	mmol/L	Minimum
mean_potassium	Mean Potassium	mmol/L	Average
max_potassium	Maximum Potassium	mmol/L	Maximum
min_bun	Minimum Blood Urea Nitrogen	mg/dL	Minimum
mean_bun	Mean Blood Urea Nitrogen	mg/dL	Average
max_bun	Maximum Blood Urea Nitrogen	mg/dL	Maximum
min_platelets	Minimum Platelet Count	K/uL	Minimum
mean_platelets	Mean Platelet Count	K/uL	Average
max_platelets	Maximum Platelet Count	K/uL	Maximum

Renal Function (Output Events)

Variable	Description	Unit	Aggregation
total_urine_72h	Total urine output in first 72 hours	mL	Sum